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Inference for L_2-Boosting

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Abstract We propose a statistical inference framework for the component-wise functional gradient descent algorithm (CFGD) under normality assumption for model errors, also known as L_2-Boosting. The CFGD is one of the most versatile tools to analyze data, because it scales well to high-dimensional data sets, allows for a very flexible definition of additive regression models and incorporates inbuilt variable selection. Due to the variable selection, we build on recent proposals for post-selection inference. However, the iterative nature of component-wise boosting, which can repeatedly select the same component to update, necessitates adaptations and extensions to existing approaches. We propose tests and confidence intervals for linear, grouped and penalized additive model components selected by L_2-Boosting. Our concepts also transfer to slow-learning algorithms more generally, and to other selection techniques which restrict the response space to more complex sets than polyhedra. We apply our framework to an additive model for sales prices of residential apartments and investigate the properties of our concepts in simulation studies.

Keywords Bootstrap · Functional Gradient Descent Boosting · Post-Selection Inference · Selective Inference · Slow Learner

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1 Introduction

We propose statistical inference techniques for the component-wise functional gradient descent algorithm (CFGD; see, e.g., Hothorn et al., 2010). CFGD emerged from the field of machine learning (c.f. Friedman, 2001), but has since also become an algorithm used to estimate statistical models (see, e.g., Mayr et al., 2017a; Melcher et al., 2017; Rügamer et al., 2018; Brockhaus et al., 2018). The CFGD is an iterative procedure, which performs model updates in the direction of the steepest decent with respect to a chosen loss function and, in contrast to other gradient boosting algorithms, only adds one single additive term (base-learner) to the model in each iteration. The algorithm is typically used in applications, where the goal is to obtain variable selection, similar to the Lasso but with the additional flexibility to estimate any type of additive regression model. The variable selection is implicitly given by the component-wise updates in combination with early stopping of the algorithm to avoid overfitting. Examples for additive regression models, which are based on the CFGD fitting procedure, are generalized additive models or functional regression models, potentially in combination with a non-normal response. In some applications such as com-

plex function-on-function regression (see, e.g., Rügamer et al., 2018), the CFGD also facilitates the estimation and modular extension of a model, which cannot be fitted with other standard software packages. The main difference and advantage lies in its component-wise fitting nature, iteratively fitting only one additive term to the response at a time and thereby allowing for a large number of covariates with manageable computational costs. A commonly used and well studied special CFGD algorithm is L_2-*Boosting* (Bühlmann and Yu, 2003). No general inferential concepts in the sense of classical statistical inference have been proposed for L_2-Boosting yet. Ad-hoc solutions such as a non-parametric bootstrap are often used to quantify the variability of boosting estimates (see e.g. Brockhaus et al., 2015; Rügamer et al., 2018), although this does not lead to confidence intervals with proper coverage. In many research areas uncertainty quantification is indispensable. We propose a framework to conduct valid inference for regression coefficients in models fitted with L_2-Boosting by conditioning on the selected covariates. We build on recent research findings on *selective inference*, which transfer classical statistical inference to algorithms with preceding selection of model terms, as is also the case for CFGD algorithms.

Standard inference is invalid after model selection, as mentioned by many authors throughout the last few decades (see, e.g., Berk et al., 2013), and a suitable inference framework is required. Different approaches for inference in high-dimensional regression models have emerged over the past few years, including data splitting (Wasserman and Roeder, 2009) and more recently, post-selection inference (PoSI; Berk et al., 2013) for valid statistical inference after arbitrary selection procedures. In this paper, classical statistical inference refers to inference concepts usually applied to assess uncertainty in regression models that do not account for a preceding selection or model choice in any sense, but treat the empirically selected model as given a priori. Invalidity of classical statistical inference methods after model selection can, in part, be explained by the fact, that the data generating process of the response will usually not yield only one specific but different selected models for a given model selection procedure for different realizations $\boldsymbol{y}$ of the response $\boldsymbol{Y} \in \mathcal{Y}$. From a geometrical point of view, different subspaces of the space $\mathcal{Y}$ will thus yield different selected models. When conditioning on a specific model for inference statements, this can be regarded as conditioning on a subspace of $\mathcal{Y}$ for inference. Classical inference methods, however, assume that the model is known prior to the analysis and hence that $\mathcal{Y}$ is not restricted. A restriction of the space of $\boldsymbol{Y}$ in turn results in a restriction of the distribution for $\hat{\boldsymbol{\beta}}$, which, if not accounted for, yields to over optimistic inference statements for the estimated parameters. We focus on *selective inference*, which provides inference statements conditional on the observed model selection. Similar to data splitting, selective inference separates the information in the data used for model selection from the information used to infer about parameters post model selection. In contrast to the original PoSI idea of providing simultaneous inference for every possible model selection, selective inference is designed to yield less conservative inference statements.

Fithian et al. (2014) have developed a general theory for selective inference in exponential family models following any type of selection mechanism. Additionally, different explicit selective inference frameworks have been derived for several selection methods (see e.g. Lee et al., 2016, for selective inference after Lasso selection or Rügamer and Greven, 2018, for selective inference after likelihood- and test-based model selection). Recent work, which we adapt and extend, aims for valid infer-

ence in forward stepwise regression (Tibshirani et al., 2016; Loftus and Taylor, 2014, 2015).

Compared to these approaches, inference for L_2-Boosting carries additional challenges due to an iterative procedure that can repeatedly select the same model term. We also extend our approach to allow for non-linear covariate effects, in contrast to existing approaches.

Our contributions are as follows: 1. We explicitly derive the space restriction of the response given by the L_2-Boosting path and thereby allow for inference as proposed in Tibshirani et al. (2016). 2. We propose a new and more powerful conditional inference concept for L_2-Boosting by conditioning only on the set of selected variables, i.e., on a set of possible selection paths. This idea can also be used for other slow learning algorithms that would require conditioning on additional quantities, with a resulting potential loss in power, to obtain an analytic representation of the inference space. For additive model structures we consider slow learners as algorithms that can repeatedly use the same additive term to gradually update a model, often by adding or deleting one covariate respectively from the model at a time. The CFGD or Forward Stagewise Regression are known examples exhibiting this behaviour. Another example is the Lasso, where an analytic representation of the inference space only becomes feasible after additionally conditioning on a list of signs and the order of variables selected. 3. We compute p-values and (two-sided) confidence intervals by Monte Carlo approximation following the results of Tibshirani et al. (2016) and Yang et al. (2016). This circumvents an explicit mathematical representation of the space the test statistic is truncated to. We refine their approach with a sampling routine that is more efficient in our setting. This approach is more generally applicable whenever the model of interest is of additive nature and the response variable is assumed to be normally distributed. 4. We extend the inference concept to account for cross-validation, stability selection (Shah and Samworth, 2013) and similar sub-sampling methods. 5. We further extend the approach to models including L_2-penalized additive effects, such as smooth effects.

Below, we summarize the L_2-Boosting algorithm in Section 2 and the concept of selective inference for sequential regression procedures in Section 3. We discuss the challenges accompanying an inference framework for L_2-Boosting and our proposed solutions in Section 4. Section 5 presents simulation results. Section 6 analyzes sales prices of real estate apartments in Tehran using our new approach. We discuss limitations and further extensions of the approach in Section 7. An add-on R-package to the model-based boosting R package `mboost` is available at `https://github.com/davidruegamer/iboost` and can be used to conduct inference for boosted models and to reproduce the results of sections 5 and 6. Further simulation and application results as well as a code to reproduce the simulation results are given in the Supplementary Material.

2 L_2-Boosting

Let $\boldsymbol{X} \in \mathbb{R}^{n \times p}$ be a fixed set of covariates and $\boldsymbol{y}$ a realization of the random response variable $\boldsymbol{Y} \in \mathbb{R}^n$. The goal of component-wise gradient boosting (see, e.g., Bühlmann and Hothorn, 2007) is to minimize a loss function $\ell(\cdot, \boldsymbol{y})$ for the given realization $\boldsymbol{y}$ with respect to an additive model $\boldsymbol{f} := \sum_{j=1}^{J} g_j(\boldsymbol{X}_j)$, where function evaluations of g_j are evaluated row-wise. The functions $g_j(\cdot)$, the so called base-learners, are defined for column subsets $\boldsymbol{X}_j \in \mathbb{R}^{n \times p_j}$ of $\boldsymbol{X}$ with $1 \leq p_j \leq p$ and can be fitted to some vector $\boldsymbol{u}^{(m)} \in \mathbb{R}^n$, which yields $\hat{\boldsymbol{g}}_j^{(m)}$ as estimate for $g_j(\boldsymbol{X}_j)$. We estimate $\boldsymbol{f}$ by $\hat{\boldsymbol{f}}$ using the component-wise functional gradient descent algorithm:

(1) Initialize an offset value $\hat{\boldsymbol{f}}^{(0)} \in \mathbb{R}^n$. If $\boldsymbol{y}$ is centered, a natural choice is $\hat{\boldsymbol{f}}^{(0)} = (0, \dots, 0)^\top$. Define $m = 0$.

(2) Do the following for $m = 1, \dots, m_{stop}$:

(2.1) Compute the pseudo-residuals $\boldsymbol{u}^{(m)} \in \mathbb{R}^n$ of step m as $\boldsymbol{u}^{(m)} = -\frac{\partial}{\partial \boldsymbol{f}} \ell(\boldsymbol{f}, \boldsymbol{y})\Big|_{\boldsymbol{f}=\hat{\boldsymbol{f}}^{(m-1)}}$.

(2.2) Approximate the negative gradient vector $\boldsymbol{u}^{(m)}$ with $\hat{\boldsymbol{g}}_j^{(m)}$ by fitting each of the base-learners $g_j(\cdot), j = 1, \dots, J$ to the pseudo-residuals and find the base-learner $j^{(m)}$, for which $j^{(m)} = \text{argmin}_{1 \leq j \leq J} ||\boldsymbol{u}^{(m)} - \hat{\boldsymbol{g}}_j^{(m)}||_2^2$ holds.

(2.3) Update $\hat{\boldsymbol{f}}^{(m)} = \hat{\boldsymbol{f}}^{(m-1)} + \nu \cdot \hat{\boldsymbol{g}}_{j^{(m)}}^{(m)}$, where $\nu \in (0, 1]$ is the so called *step-length* or *learning rate* and usually fixed to some sufficiently small value such as 0.1 or 0.01 (Bühlmann and Hothorn, 2007).

When defining $\ell(\boldsymbol{f}, \boldsymbol{y}) = \frac{1}{2}||\boldsymbol{y} - \boldsymbol{f}||_2^2$ with quadratic L_2-Norm $||\cdot||_2^2$, L_2-Boosting is obtained, which corresponds to mean regression using the model $\mathbb{E}(\boldsymbol{Y}|\boldsymbol{X}) = \sum_{j=1}^J g_j(\boldsymbol{X}_j)$. The vector $\boldsymbol{u}^{(m)}$ then corresponds to the residuals $\boldsymbol{y} - \hat{\boldsymbol{f}}^{(m-1)}$. In the framework of additive regression models, each base-learner $g_j(\cdot)$ constitutes a partial effect and is represented as a linear effect of a covariate or of a basis evaluated at that covariate vector, i.e., $\boldsymbol{g}_j(\boldsymbol{X}_j) = \boldsymbol{X}_j\boldsymbol{\beta}_j$. The coefficient $\boldsymbol{\beta}_j$ is estimated using ordinary or penalized least squares. The model fit $\hat{\boldsymbol{g}}_j^{(m)}$ of each base-learner in the mth step is therefore given by $\hat{\boldsymbol{g}}_j^{(m)} = \boldsymbol{H}_j\boldsymbol{u}^{(m)} = \boldsymbol{X}_j(\boldsymbol{X}_j^\top\boldsymbol{X}_j + \lambda_j\boldsymbol{D}_j)^{-1}\boldsymbol{X}_j^\top\boldsymbol{u}^{(m)}$, where the hat matrix $\boldsymbol{H}_j$ is defined by the corresponding design matrix $\boldsymbol{X}_j$, a penalty matrix $\boldsymbol{D}_j$ and a pre-specified smoothing parameter $\lambda_j \geq 0$ controlling the penalization. As only one base-learner is chosen in each iteration, the final effective degrees of freedom of the jth base-learner depend on the number of selections.

L_2-Boosting scales well to large data sets due to its component-wise fitting nature and is particularly suited for the estimation of structured additive regression models. It has the additional advantage of being able to handle $n < p$-settings and conducting variable selection, as not all J model terms are necessarily selected in at least one iteration. However, variable selection has to be accounted for when constructing uncertainty measures for regression coefficients, as it restricts the space of $\boldsymbol{Y}$ and thus of the estimated parameters.

3 Selective Inference

3.1 Considered Setup

Let $\boldsymbol{Y} = \boldsymbol{\mu} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2\boldsymbol{I}_n)$ and n-dimensional identity matrix $\boldsymbol{I}_n$. Furthermore, assume that σ^2 is known and $\boldsymbol{\mu}$ is an unknown parameter of interest. We do not assume any true linear relationship between $\boldsymbol{\mu}$ and covariates, but estimate $\boldsymbol{\mu}$ with an additive "working model" based on fixed covariates $\boldsymbol{X} \in \mathbb{R}^{n \times p}$ with p potentially exceeding n. Furthermore, define the selection procedure or selection event

$$\mathcal{S} : \mathbb{R}^n \to \mathcal{P}(\{1, \dots, p\}), \boldsymbol{y} \mapsto \mathcal{S}(\boldsymbol{y})$$

with power set function $\mathcal{P}(\cdot)$. For the given realization $\boldsymbol{y}$ of $\boldsymbol{Y}$, we denote $\mathcal{S}(\boldsymbol{y}) =: \mathcal{A}$, for which we assume $|\mathcal{A}| \leq n$.

We focus on estimating the best linear projection of $\boldsymbol{\mu}$ into the space spanned by the variables given by $\mathcal{A}$ after model selection and making uncertainty statements about any direction of this projection, i.e., the significance of any linear covariate effect, given the selected model $\mathcal{A}$. We therefore run the selection procedure defined by $\mathcal{S}$, select the subset $\boldsymbol{X}_\mathcal{A}$ of $\boldsymbol{X}$ defined by the selected column indices $\mathcal{S}(\boldsymbol{y}) = \mathcal{A}$ and estimate regression coefficients $\boldsymbol{\beta}_\mathcal{A}$ by projecting $\boldsymbol{y}$ into the linear subspace $\boldsymbol{W}_\mathcal{A} \subseteq \mathbb{R}^n$ spanned by the columns of $\boldsymbol{X}_\mathcal{A}$. Our inference goal is to test one entry β_j in $\boldsymbol{\beta}_\mathcal{A}$, i.e.,

$$H_0 : \beta_j = \beta_{j,0},$$

conditional on the selected model, which is equivalent to testing

$$H_0 : \boldsymbol{v}^\top \boldsymbol{\mu} := \boldsymbol{e}_j^T (\boldsymbol{X}_\mathcal{A}^\top \boldsymbol{X}_\mathcal{A})^{-1} \boldsymbol{X}_\mathcal{A}^\top \boldsymbol{\mu} = \beta_{j,0} \tag{1}$$

with $\boldsymbol{e}_j$ the unit vector selecting $j \in \mathcal{A}$ (see, e.g., Tibshirani et al., 2016). Without selection, (1) can be tested using $\tilde{R} := \boldsymbol{v}^\top \boldsymbol{Y}$, which follows a normal distribution with expectation $\tilde{\rho} = \boldsymbol{v}^\top \boldsymbol{\mu}$ and variance $\sigma^2 \boldsymbol{v}^\top \boldsymbol{v}$ under the null. However, after model selection, the space of $\boldsymbol{Y}$ is restricted to $\mathcal{G} = \{\boldsymbol{y} : \mathcal{S}(\boldsymbol{y}) = \mathcal{A}\}$, which we call the *inference region*. Many of the proposed methods for selective inference then describe this space restriction mathematically and derive the distribution of $\boldsymbol{v}^\top \boldsymbol{Y} \,|\, \boldsymbol{Y} \in \mathcal{G}$.

Let $\boldsymbol{P_W}$ generally be the projection onto a linear subspace $\text{span}(\boldsymbol{W}) \subset \mathbb{R}^n$ defined by some $\boldsymbol{W} \in \mathbb{R}^{n \times w}$, $w \in \mathbb{N}$, and $\boldsymbol{P}_{\boldsymbol{W}}^\perp$ be the projection onto the orthogonal complement of this linear subspace. Furthermore, define the direction of $\boldsymbol{P_W y}$ as the unit vector $\text{dir}_{\boldsymbol{W}}(\boldsymbol{y}) = \frac{\boldsymbol{P_W y}}{||\boldsymbol{P_W y}||_2}$.

We now shortly review three approaches to selective inference derived for a similar setup and build on these ideas in Section 4.

3.2 Existing Approaches for Other Procedures

For sequential regression procedures such as Forward Stepwise Regression (FSR) or the Least Angle Regression (LAR, Efron et al., 2004), Tibshirani et al. (2016) characterize the restricted region of the on-going selection mechanism as a polyhedral set $\mathcal{G} = \{\boldsymbol{y} : \boldsymbol{\Gamma y} \geq \boldsymbol{b}\}$ with $\boldsymbol{\Gamma} \in \mathbb{R}^{\iota \times n}$, $\boldsymbol{b} \in \mathbb{R}^\iota$ for some $\iota \in \mathbb{N}$ and an inequality $\geq$ which is to be interpreted componentwise.

By additionally conditioning on the realization $\boldsymbol{z}$ of $\boldsymbol{Z} = \boldsymbol{P}_{\boldsymbol{v}}^\perp \boldsymbol{Y}$ as well as on a list of signs for each step similar to those defined in (9) and which will be explained in Section 4, $\tilde{R}$ follows a truncated Gaussian distribution $\mathcal{N}(\tilde{\rho}, \sigma^2 \boldsymbol{v}^\top \boldsymbol{v})$ with analytically describable truncation limits $\mathcal{V}^{lo} = \mathcal{V}^{lo}(\boldsymbol{z}), \mathcal{V}^{up} = \mathcal{V}^{up}(\boldsymbol{z})$ (see Lee et al., 2016). Let $F_{\tilde{\rho}, \sigma^2 \boldsymbol{v}^\top \boldsymbol{v}}^{[\mathcal{V}^{lo}, \mathcal{V}^{up}]}(\tilde{R})$ denote the cumulative distribution function of this truncated normal distribution evaluated at $\tilde{R}$. Then, for $H_0 : \tilde{\rho} \leq 0$ vs. $H_1 : \tilde{\rho} > 0$, the test statistic $T = 1 - F_{0, \sigma^2 \boldsymbol{v}^\top \boldsymbol{v}}^{[\mathcal{V}^{lo}, \mathcal{V}^{up}]}(\tilde{R})$ is a valid conditional p-value, conditional on the polyhedral selection, as $\mathbb{P}_{H_0}(T \leq \alpha \mid \boldsymbol{\Gamma Y} \geq \boldsymbol{b}) = \alpha$ for any $0 \leq \alpha \leq 1$. Two-sided p-values can be constructed as $T = 2 \cdot \min\left(F_{0, \sigma^2 \boldsymbol{v}^\top \boldsymbol{v}}^{[\mathcal{V}^{lo}, \mathcal{V}^{up}]}(\tilde{R}), 1 - F_{0, \sigma^2 \boldsymbol{v}^\top \boldsymbol{v}}^{[\mathcal{V}^{lo}, \mathcal{V}^{up}]}(\tilde{R})\right)$ (Tibshirani et al., 2016).

The characterization of the inference region as a polyhedral set, however, is only possible if the algorithmic decision in each selection step is a linear restriction on the space of $\boldsymbol{Y}$. Loftus and Taylor (2015) introduce a framework for inference after model selection procedures which can be described by affine inequalities, focusing on groups of variables.

For testing the jth group variable coefficient $\boldsymbol{\beta}_{\mathcal{A},j} \in \mathbb{R}^{p_j}$ in the best linear approximation $\boldsymbol{\beta}_\mathcal{A} = \arg\min \mathbb{E}[\,||\boldsymbol{Y} - \boldsymbol{X}_\mathcal{A}\boldsymbol{\beta}||_2^2\,]$, Loftus and Taylor (2015); Yang et al. (2016) rewrite the null hypothesis $\boldsymbol{\beta}_{\mathcal{A},j} = \boldsymbol{0}$ as $\boldsymbol{P_W \mu} = \boldsymbol{0}$ or

$$H_0 : \rho := ||\boldsymbol{P_W \mu}||_2 = 0 \tag{2}$$

with $\boldsymbol{W} = \boldsymbol{P}_{\boldsymbol{X}_{\mathcal{A}\setminus j}}^\perp \boldsymbol{X}_j$, where $\boldsymbol{X}_{\mathcal{A}\setminus j}$ denotes $\boldsymbol{X}_\mathcal{A}$ without the p_j columns corresponding to the jth group variable. Under the null and when additionally conditioning on the direction $\text{dir}_{\boldsymbol{W}}(\boldsymbol{y})$, $R := ||\boldsymbol{P_W Y}||_2$ follows a truncated χ-distribution with analytically derivable limits. Yang et al. (2016) note that R and $\text{dir}_{\boldsymbol{W}}(\boldsymbol{y})$ are not independent for $\rho \neq 0$ and as a consequence, the χ-conditional distribution of R as derived in Loftus and Taylor (2015) for (2) when $\rho = 0$ no longer holds for more general hypotheses, as relevant for the derivation of confidence intervals.

Yang et al. (2016) decompose $\boldsymbol{Y}$ as $R \cdot \text{dir}_{\boldsymbol{W}}(\boldsymbol{Y}) + \boldsymbol{P}^{\perp}_{\boldsymbol{W}}\boldsymbol{Y}$ and condition on $\text{dir}_{\boldsymbol{W}}(\boldsymbol{Y}) = \text{dir}_{\boldsymbol{W}}(\boldsymbol{y})$ as well as on $\boldsymbol{P}^{\perp}_{\boldsymbol{W}}\boldsymbol{Y} = \boldsymbol{P}^{\perp}_{\boldsymbol{W}}\boldsymbol{y}$. Then, the only variation left is in R and the selection $\mathcal{A}$ can be equally written as $R \in \mathcal{R}_y$ with

$$\mathcal{R}_y = \left\{R > 0 : \mathcal{S}(R \cdot \text{dir}_{\boldsymbol{W}}(\boldsymbol{y}) + \boldsymbol{P}^{\perp}_{\boldsymbol{W}}\boldsymbol{y}) = \mathcal{A}\right\}. \tag{3}$$

The distribution of R conditional on the selection, on $\text{dir}_{\boldsymbol{W}}(\boldsymbol{y})$ as well as on $\boldsymbol{P}^{\perp}_{\boldsymbol{W}}\boldsymbol{y}$, has a density proportional to

$$R^{w-1} \exp\left\{-\frac{1}{2\sigma^2}(R^2 - 2R \cdot \langle \text{dir}_{\boldsymbol{W}}(\boldsymbol{y}), \boldsymbol{\mu}\rangle)\right\} \cdot \mathbb{1}\{R \in \mathcal{R}_y\} \tag{4}$$

with indicator function $\mathbb{1}\{\cdot\}$. (4) can be used to conduct inference on the inner product $\langle \text{dir}_{\boldsymbol{W}}(\boldsymbol{y}), \boldsymbol{\mu}\rangle$. As $\rho = ||\boldsymbol{P}_{\boldsymbol{W}}\boldsymbol{\mu}||_2 \geq \langle \text{dir}_{\boldsymbol{W}}(\boldsymbol{y}), \boldsymbol{\mu}\rangle$ holds, (4) can also be used to construct a lower bound for the quantity of interest ρ.

An explicit definition of the inference region is, however, not necessary. Theorem 1 in Yang et al. (2016) states that, conditional on $\text{dir}_{\boldsymbol{W}}(\boldsymbol{y})$, $\boldsymbol{P}^{\perp}_{\boldsymbol{W}}\boldsymbol{y}$ and the selection event, inference can be conducted using the Uniform$[0, 1]$ p-value $\varsigma(t_y)$ for $H_0 : \langle \text{dir}_{\boldsymbol{W}}(\boldsymbol{y}), \boldsymbol{\mu}\rangle = t_y$ with

$$\varsigma(t) = \frac{\int_{R \in \mathcal{R}_y, R > ||\boldsymbol{P}_{\boldsymbol{W}}\boldsymbol{y}||_2} R^{w-1} e^{-(R^2 - 2Rt)/2\sigma^2} \, \mathrm{d}R}{\int_{R \in \mathcal{R}_y} R^{w-1} e^{-(R^2 - 2Rt)/2\sigma^2} \, \mathrm{d}R}. \tag{5}$$

The authors note that (5) is equal to

$$\frac{\mathbb{E}_{R \sim \sigma\chi_w}(e^{Rt/\sigma^2} \cdot \mathbb{1}\{R \in \mathcal{R}_y, R > ||\boldsymbol{P}_{\boldsymbol{W}}\boldsymbol{y}||_2\})}{\mathbb{E}_{R \sim \sigma\chi_w}(e^{Rt/\sigma^2} \cdot \mathbb{1}\{R \in \mathcal{R}_y\})}, \tag{6}$$

which can be approximated by the ratio of empirical expectations computed with a large number of samples $r^b \sim \sigma \cdot \chi_w, b = 1, \ldots, B$. To evaluate the argument of both expectations in (6) for some r^b, $r^b \in \mathcal{R}_y$ must be checked. Note that the only variation of $(\boldsymbol{Y} \mid \text{dir}_{\boldsymbol{W}}(\boldsymbol{y}), \boldsymbol{P}^{\perp}_{\boldsymbol{W}}\boldsymbol{y})$ is in R. Therefore, define $\boldsymbol{y}^b = \boldsymbol{P}^{\perp}_{\boldsymbol{W}}\boldsymbol{y} + r^b \cdot \text{dir}_{\boldsymbol{W}}(\boldsymbol{y})$ and rerun the algorithm to check whether $\mathcal{S}(\boldsymbol{y}^b) = \mathcal{A}$, or equivalently, whether $r^b \in \mathcal{R}_y$. Drawing samples from the $\sigma\chi_w$-distribution is inefficient, however, when $||\boldsymbol{P}_{\boldsymbol{W}}\boldsymbol{y}||_2$ is far away from the null as then an excessively large number of samples is needed to obtain a good approximation of $\varsigma(t)$. Yang et al. (2016) therefore suggest an importance sampling algorithm, which draws samples r^b from a proposal distribution $\mathcal{F}_{prop}$ such as $\mathcal{N}(||\boldsymbol{P}_{\boldsymbol{W}}\boldsymbol{y}||_2, \sigma^2)$ with density f_{prop} and then approximates (6) by

$$\varsigma(t) \approx \hat{\varsigma}(t) = \frac{\sum_b w_b \cdot e^{r^b t/\sigma^2} \cdot \mathbb{1}\{r^b \in \mathcal{R}_Y, r^b > ||\boldsymbol{P}_{\boldsymbol{W}}\boldsymbol{y}||_2\}}{\sum_b w_b \cdot e^{r^b t/\sigma^2} \cdot \mathbb{1}\{r^b \in \mathcal{R}_Y\}} \tag{7}$$

with sampling weights $w_b = f_{\sigma\chi_w}(r^b)/f_{prop}(r^b)$.

4 Selective Inference concepts for L_2-Boosting

We now propose selective inference concepts for L_2-Boosting. In Section 4.1 we first derive a polyhedron representation of selection conditions in L_2-Boosting. After discussing the resulting inference framework based on existing concepts and its lack of power in Section 4.2, we propose an alternative concept for L_2-Boosting and similar slow learners, which can repeatedly select the same base-learners. Based on this idea, we derive a powerful inference framework for L_2-Boosting with linear base-learners in Section 4.3 and describe important extensions in Section 4.4.

4.1 Polyhedron representation-based inference for L_2-Boosting

Consider L_2-Boosting using only linear base-learners, i.e., $\boldsymbol{D}_j = \boldsymbol{0}, \boldsymbol{X}_j \in \mathbb{R}^{n \times 1} \, \forall \, j$. Similar to Tibshirani et al. (2016), we can derive a polyhedron representation $\mathcal{G} = \{\boldsymbol{y} : \boldsymbol{\Gamma}\boldsymbol{y} \geq \boldsymbol{b}\}$ **for the given selection path** $j^{(1)}, \ldots, j^{(m_{\text{stop}})}$ of L_2-Boosting.

The selection condition for the mth chosen base-learner

$$\begin{aligned} & ||(\boldsymbol{I} - \boldsymbol{H}_{j^{(m)}})\boldsymbol{u}^{(m)}||^2 \leq ||(\boldsymbol{I} - \boldsymbol{H}_j)\boldsymbol{u}^{(m)}||^2 \\ \Leftrightarrow \quad & \left(s_m \boldsymbol{X}_{j^{(m)}}^\top / ||\boldsymbol{X}_{j^{(m)}}||_2 \pm \boldsymbol{X}_j^\top / ||\boldsymbol{X}_j||_2\right) \boldsymbol{u}^{(m)} \geq 0, \end{aligned} \tag{8}$$

which holds $\forall j \neq j^{(m)}$ with $s_m = \text{sign}(\boldsymbol{X}_{j^{(m)}}^\top \boldsymbol{u}^{(m)})$, can be written as affine restriction on $\boldsymbol{y}$ by plugging the residual vector $\boldsymbol{u}^{(m)}$ of step m as a function of $\boldsymbol{y}$

$$\boldsymbol{u}^{(m)} = \left[\prod_{l=1}^{m-1} \left(\boldsymbol{I} - \nu \boldsymbol{H}_{j^{(m-l)}}\right)\right] \boldsymbol{y} =: \Upsilon^{(m)} \boldsymbol{y}$$

into (8). For a given selection path and list of signs $s_m, m = 1, \ldots, m_{\text{stop}}$ this yields the polyhedron representation $\mathcal{G}$ with fixed $(2 \cdot (p-1) \cdot m_{\text{stop}}) \times n$ matrix $\boldsymbol{\Gamma}$ as stacked matrix of n-dimensional row vectors, where the rows $(\tilde{m} + 2(j - \omega(j)) - 1)$ and $(\tilde{m} + 2(j - \omega(j)))$ of $\boldsymbol{\Gamma}$ with $\tilde{m} = 2 \cdot (p-1) \cdot (m-1)$ and $\omega(j) = \mathbb{1}\{j > j^{(m)}\}$ are given by

$$\left(s_m \boldsymbol{X}_{j^{(m)}}^\top / ||\boldsymbol{X}_{j^{(m)}}||_2 \pm \boldsymbol{X}_j^\top / ||\boldsymbol{X}_j||_2\right) \Upsilon^{(m)} \quad \forall\, j \neq j^{(m)}. \tag{9}$$

As for other procedures described in the post-selection inference literature, this representation only holds if the columns of $\boldsymbol{X}$ are in general position, which however, is not a very stringent assumption (see, e.g., Tibshirani et al., 2016, Section 4).

As the L_2-Boosting path results in a polyhedral set as space restriction for $\boldsymbol{Y}$, conditional on the list of signs, quantities of interest $\boldsymbol{v}^\top \boldsymbol{\mu}$ can be tested based on the conditional distribution of $\boldsymbol{v}^\top \boldsymbol{Y} \,|\, \boldsymbol{Y} \in \mathcal{G}$ as proposed by Tibshirani et al. (2016). To this end, we have to condition on the selection path. If we do not additionally condition on the list of signs, $\mathcal{G}$ is a union of polyhedra (cf. Lee et al., 2016).

If group base-learners or base-learners with penalties are used, space restrictions no longer yield a polyhedron. Instead, affine inequalities can be used to obtain truncation limits analogous to Loftus and Taylor (2015); Rügamer and Greven (2018).

4.2 Choice of the Conditioning Event for Slow Learners

For the selection approaches discussed in Section 3, conditioning on the selection path helps to derive the corresponding conditional distribution and, compared to conditioning on the selected model only, additionally conditions on the selection order of variables and their effect sign. For boosting and other slow learners that can repeatedly select the same base-learner, conditioning on the selection path and thus on variable selection decisions in each algorithmic step will result in an even larger loss of power. In fact, such a conditional inference will have almost no power in most practically relevant situations, as we show empirically for the polyhedron approach in the simulation section. In order to avoid excessive conditioning, we propose conditioning only on the set of selected covariates (not on the selection order or the effect signs), i.e., on the selected statistical model.

Conditioning only on the selected covariates, however, means that the mathematical description of the inference region becomes far more difficult. For L_2-Boosting with linear base-learners, this would result in a union of not necessarily overlapping polyhedra for the different selection paths leading to the same selected model. We do not think that a general analytical description of this inference region is possible. We thus circumvent this problem by using a Monte Carlo approximation, adapting and extending the existing approaches summarized in Section 3.2.

4.3 Powerful Inference for L_2-Boosting with Linear Base-learners

We base inference on the potentially multiply truncated Gaussian distribution of $\tilde{R} = \boldsymbol{v}^\top \boldsymbol{Y}$ conditional on $\boldsymbol{P}_v^\perp \boldsymbol{y}$ and the selection $\tilde{R} \in \mathcal{R}_y$. Then, the truncated normal density of $\tilde{R}$ is given by

$$f(R) \propto \exp\left\{-\frac{1}{2\sigma^2 \boldsymbol{v}^\top \boldsymbol{v}}(R - \boldsymbol{v}^\top \boldsymbol{\mu})^2\right\} \cdot \mathbb{1}\{R \in \mathcal{R}_y\}, \tag{10}$$

where $\mathcal{R}_y$ is a union of polyhedra. The proof of equation (10) follows analogously to Lemma 1 of Yang et al. (2016) for $\tilde{R} = \boldsymbol{v}^\top \boldsymbol{Y}$ using $w = 1$ (cf. (4)). Note that $\langle \text{dir}_{\boldsymbol{W}}(\boldsymbol{y}), \boldsymbol{\mu} \rangle = \boldsymbol{v}^\top \boldsymbol{\mu} / ||\boldsymbol{v}||_2$ in this case; we rescaled $\tilde{R}$ compared to the definition before and kept the sign by using a normal instead of a χ-distribution. Let $r_{\text{obs}} = \boldsymbol{v}^\top \boldsymbol{y}$. Then, analogous to Yang et al. (2016) we can define a p-value by

$$\varsigma(\beta_{j,0}) = \frac{\int_{\tilde{R} > r_{\text{obs}}, \tilde{R} \in \mathcal{R}_y} e^{-(2\sigma^2 \boldsymbol{v}^\top \boldsymbol{v})^{-1}(\tilde{R}^2 - 2\tilde{R}\beta_{j,0})} \, \mathrm{d}\tilde{R}}{\int_{\tilde{R} \in \mathcal{R}_y} e^{-(2\sigma^2 \boldsymbol{v}^\top \boldsymbol{v})^{-1}(\tilde{R}^2 - 2\tilde{R}\beta_{j,0})} \, \mathrm{d}\tilde{R}}$$

for $H_0 : \boldsymbol{v}^\top \boldsymbol{\mu} = \beta_{j,0}$ and since the truncated Gaussian distribution with potentially multiple truncation limits increases monotonically in its mean ρ (see, e.g., Rügamer and Greven, 2018), we can find unique values $\rho_{\alpha/2}, \rho_{1-\alpha/2}$ for any $\alpha \in (0, 1)$, such that

$$\varsigma(\rho_a) = \frac{\int_{\tilde{R} > r_{\text{obs}}, \tilde{R} \in \mathcal{R}_y} e^{-(2\sigma^2 \boldsymbol{v}^\top \boldsymbol{v})^{-1}(\tilde{R}^2 - 2\tilde{R}\rho_a)} \, \mathrm{d}\tilde{R}}{\int_{\tilde{R} \in \mathcal{R}_y} e^{-(2\sigma^2 \boldsymbol{v}^\top \boldsymbol{v})^{-1}(\tilde{R}^2 - 2\tilde{R}\rho_a)} \, \mathrm{d}\tilde{R}} = a,$$

$a \in \{\alpha/2, 1 - \alpha/2\}$, to construct a two-sided confidence interval $[\rho_{\alpha/2}, \rho_{1-\alpha/2}]$. This is an extension of the one-sided confidence intervals of Yang et al. (2016).

Note that $\varsigma(\rho_a)$ can then be rewritten as

$$\frac{\mathbb{E}_{\tilde{R} \sim \mathcal{N}(0, \sigma^2 \boldsymbol{v}^\top \boldsymbol{v})}\left[\mathbb{1}\{\tilde{R} \in \mathcal{R}_y, \tilde{R} > r_{\text{obs}}\} \cdot e^{(\sigma^2 \boldsymbol{v}^\top \boldsymbol{v})^{-1} \tilde{R} \rho_a}\right]}{\mathbb{E}_{\tilde{R} \sim \mathcal{N}(0, \sigma^2 \boldsymbol{v}^\top \boldsymbol{v})}\left[\mathbb{1}\{\tilde{R} \in \mathcal{R}_y\} \cdot e^{(\sigma^2 \boldsymbol{v}^\top \boldsymbol{v})^{-1} \tilde{R} \rho_a}\right]}, \tag{11}$$

which allows for an empirical approximation as in (7). Further note that this approach does not require to condition on the list of signs nor on the order of the selected variables. As this has been necessary to obtain selective inference statements for existing approaches such as the Lasso (Lee et al., 2016), our method can also be used to condition on less for these approaches and thus potentially leads to more powerful inference.

Monte Carlo Approximation

In practice, importance sampling from $\mathit{\Pi} = \mathcal{N}(r_{\text{obs}}, \sigma^2 \boldsymbol{v}^\top \boldsymbol{v})$ approximates (11) well if the given truncation limits around r_{obs} are fairly symmetric, yielding the weights $w_b = \exp((2r^b r_{\text{obs}} - r_{\text{obs}}^2) \cdot (-2\sigma^2 \boldsymbol{v}^\top \boldsymbol{v})^{-1})$ for the importance sampler. A refinement of the sampling routine is necessary to also work well in more extreme cases. An example frequently encountered in practice is when r_{obs} is rather large and at the same time lies very close to one truncation limit, yielding an insufficient number of samples $r^b \in \mathcal{R}_y$ to approximate the tail of the truncated distribution well. We therefore propose a more efficient sampling routine, motivated by and applicable to selection procedures, for which the support of the truncated distribution is known to be a single interval $[\mathcal{V}^{lo}, \mathcal{V}^{up}]$. Our idea is that, in this case, we do not need to characterize the space empirically since the distribution of interest is known with the exception of the interval limits (the variance is assumed to be known and the null distribution determines the mean ρ). By employing a line search, we can find $\mathcal{V}^{lo}, \mathcal{V}^{up}$ and conduct inference based on the truncated normal distribution function $F^{[\mathcal{V}^{lo}, \mathcal{V}^{up}]}_{\rho, \sigma^2 \boldsymbol{v}^\top \boldsymbol{v}}(\cdot)$. We use such a corresponding line search here to refine the importance sampling. To find a super set of $\mathcal{R}_y$, we start with extremely small, or respectively, large quantiles R of $\mathit{\Pi} = \mathcal{N}(\rho, \sigma^2 \boldsymbol{v}^\top \boldsymbol{v})$ and check for selection congruency, i.e., whether $R \in \mathcal{R}_y$. We successively increase, or respectively, decrease the quantiles for which we perform